Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

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Abstract

Does asking a large language model to think step by step help it reason about time, and does the answer change across languages? We study this with Sarvam-105B on the Temporal Reasoning Dataset (TRD), comparing direct answers with chain-of-thought (CoT) prompting in English and Hindi, with and without controlled distractor sentences. On 3,089 questions answered in all twelve conditions, CoT raises English accuracy by 27.9 percentage points but Hindi accuracy by only 5.1, and it significantly lowers Hindi accuracy on two tasks. The English-Hindi gap under CoT widens with difficulty, and distractors make CoT more fragile only in Hindi, whose reasoning traces often weave the distractor into the solution. Hindi traces stay in Hindi, are shorter, rarely check their work, and fail on different questions from English traces. We describe the problem, related work, our pipeline, the progress so far, and a timeline for the remaining conditions: Hinglish reasoning, cross-lingual prompting, and a structural comparison of English and Hindi traces.

1 Introduction

Chain-of-thought (CoT) prompting asks a large language model (LLM) to write out intermediate steps before giving an answer. Surveys describe CoT as a way to break a difficult problem into smaller pieces, and asking for explicit steps can improve multi-step reasoning (Wei et al., 2022; Huang and Chang, 2023). We use "reasoning trace" to mean the text the model generates, and we do not assume that this text is a transparent record of its internal computation (Turpin et al., 2023).

The language matters too. The MGSM study found that multilingual models can solve reasoning problems in many languages, but English intermediate steps often perform as well as or better than steps written in the language of the question (Shi et al., 2023b). That leads to a practical question: if we ask the model to reason in Hindi, does it reason as well as it does in English, and does it get there the same way?

We study this question on temporal reasoning, using the Temporal Reasoning Dataset (TRD) of Mazzia et al. (2026). TRD generates date and time problems in ten languages at controlled difficulty, and can prepend a distractor sentence to any question. Some distractors relate to time ("I always get confused with months that have 31 days"), while others have nothing to do with the question, such as a sentence about someone buying a motorcycle. This gives us a direct test of our main idea:

> If a model can ignore the motorcycle when it answers directly, does making it think step by step give the motorcycle more chances to enter its reasoning and derail the answer?

TRD evaluates its benchmark without CoT, telling the model not to add explanations, and explicitly leaves the effect of CoT on temporal reasoning for future work (Mazzia et al., 2026). We investigate that missing condition, in English and Hindi, with Sarvam-105B (Sarvam AI, 2026).

2 Problem Statement

We work with one model, Sarvam-105B, and TRD problems whose answers have a fixed, machine-readable form: a date (YYYY-MM-DD), a clock time (HH:MM), a whole number of days or minutes, a pair of dates for an interval, or a weekday name. Every gold answer has a known form, so responses can be scored automatically.

We vary four things. The question is given in English or Hindi. The distractor is absent, a sentence about time, or a sentence about something unrelated. The prompt is *direct*, which asks only for the answer, or *CoT*, which asks the model to reason step by step; a third prompt, the cross-lingual

prompting (CLP) of Qin et al. (2023), is planned. Whenever we ask for step-by-step reasoning we also choose the language the model should reason in: so far the question's own language, and in the remaining work Hinglish as well. Running the model on one problem gives back the final answer and, in every condition except direct, a reasoning trace.

From the answers we read off differences in accuracy. The *CoT delta* is the change in accuracy when CoT replaces direct answering; a positive delta means step-by-step reasoning helped. The *distraction drop* is how much accuracy falls when a distractor is added. The quantity we care about most combines the two: the *CoT distraction penalty*, the distraction drop under CoT minus the distraction drop under direct answering. If it is positive, thinking out loud gives the distractor room to creep into the answer. We also track the *English–Hindi gap*, English accuracy minus Hindi accuracy, at each difficulty level.

The traces let us look past the final answer. We record how much of a Hindi trace stays in Hindi with the target-language ratio (TLR; Section 4), whether the trace mentions the distractor, and how its final answer relates to its reasoning. In the remaining work we will turn each trace into a short sequence of step types, such as identify a date, add a duration, or check a constraint, and compare the English and Hindi sequences for the same problem. The *reasoning divergence score* (RDS) is the normalised edit distance between the two sequences, so that 0 means the same structure and 1 means they share nothing. A logistic regression that predicts whether an answer is wrong from its RDS, difficulty and language will test whether taking a different route goes with more errors.

We ask four questions. (1) Does CoT help, and does that depend on language and difficulty? (2) Does CoT make distractors more harmful, and does that differ by distractor type or language? (3) How does reasoning in Hinglish change accuracy and trace structure? (4) Does CLP beat Hindi CoT and bring Hindi reasoning closer to English? This report answers the first two.

3 Related Work

CoT, and when it stops helping. Wei et al. (2022) introduced CoT prompting by adding worked examples whose solutions show a chain of natural-language steps; gains appeared mainly in very large models such as PaLM 540B. Kojima et al. (2022) then showed that the examples are not needed: appending "Let's think step by step" and running a second prompt to extract the answer raised text-davinci-002 from 17.7% to 78.7% on MultiArith and from 10.4% to 40.7% on GSM8K. We use this zero-shot form, since TRD has no worked solutions. Wang et al. (2023) kept the CoT prompt but sampled several paths and took a majority vote, improving GSM8K by 17.9%; CLP's ensemble variant reuses that idea. Sprague et al. (2025) combined a meta-analysis of over 100 papers with their own runs on 20 datasets and 14 models, and found that CoT helps mainly on math and symbolic problems, with most of the benefit coming from executing symbolic steps rather than planning them. Liu et al. (2025) took six tasks where thinking out loud makes humans worse and showed that models degrade in the same way, with o1-preview losing up to 36.3% against GPT-4o answering directly. TRD's reasoning tasks are date arithmetic, so the first paper predicts that CoT should help there, while its memorisation tasks look more like recall. Neither lets us assume that CoT always helps.

Distractors and irrelevant context. Shi et al. (2023a) built GSM-IC by adding one irrelevant sentence to GSM8K problems, varying its topic, whether it reused a name, and which numbers it carried. Accuracy fell sharply under CoT, zero-shot CoT, least-to-most and program prompting, and self-consistency and an explicit instruction to ignore irrelevant information both helped. Mirzadeh et al. (2025) turned GSM8K into symbolic templates so that names and numbers could be resampled, and found accuracy moving between versions of the same question and falling as clauses were added; their GSM-NoOp clause, which looks relevant but changes nothing, cost up to 65%. Neither compares CoT on and off under the same distractor, and both work in English.

Reasoning across languages. Shi et al. (2023b) built MGSM by translating 250 GSM8K problems into ten languages and comparing direct answers, native-language CoT, English CoT and translate-test; English CoT usually matched or beat native CoT. Hindi is not in MGSM, which is one reason we work with TRD. Qin et al. (2023) proposed cross-lingual prompting (CLP): a first stage asks the model to act as an expert in multilingual understanding and to work through the request

step by step in a target language, usually English, and a second stage asks it to solve the task from its own restatement. Cross-lingual self-consistent prompting (CLSP) runs the first stage in several languages and votes over the answers. Huang et al. (2023) packed the same idea into one cross-lingual-thought template and narrowed the gap between languages. Lai and Nissim (2024) found that open models do not answer the same math problem consistently across languages and instruction-tuned a 7B model on about 6.3M CoT samples in eleven languages, though Hindi is not among them. For English-centric reasoning models, Yong et al. (2025) found that more test-time thinking improves multilingual math enough to beat models twice their size, with traces that stay mostly English and quote the non-English input; models reason better in high-resource languages.

Language drift, temporal benchmarks, and trace comparison. Models seem to pivot through English internally: Llama-2's middle layers decode more readily to English than to the input language (Wendler et al., 2024), and Zhao et al. (2024) describe an understand–solve–respond workflow supported by language-specific neurons. On the output side, Marchisio et al. (2024) counted how often models answer in the wrong language and found serious confusion in several strong models, and Qi et al. (2025) saw the same in reasoning traces: leading models fall back to English, and prompts that force the user's language improve readability but cost accuracy. Our target-language ratio measures whether the model's reasoning stayed in the given language or not. Temporal reasoning has its own benchmarks, including TimeQA, TempReason, MenatQA, TRAM, TimeBench and Test of Time (Chen et al., 2021; Tan et al., 2023; Wei et al., 2023; Wang and Zhao, 2024; Chu et al., 2024; Fatemi et al., 2025), but they are English-only and none uses controlled distractors. Code-mixing has benchmarks, including Hindi–English ones (Khanuja et al., 2020; Aguilar et al., 2020; Yang and Chai, 2025), and code-mixed few-shot examples add to CoT gains (Shankar et al., 2024), but we found no study of CoT traces written in Hinglish, so we keep that condition exploratory. Finally, ROSCOE and ReCEval score reasoning chains step by step (Golovneva et al., 2023; Prasad et al., 2023), but both lean on English text models, so instead of scoring quality we will reduce each step to a language-independent type and compare sequences with edit distance (Levenshtein, 1966).

Work	Multi.	Temp.	CoT	Distr.	Trace
Wei et al. (2022)	–	○	✓	–	–
Sprague et al. (2025)	–	–	✓	–	–
Shi et al. (2023a)	–	–	✓	✓	–
Mirzadeh et al. (2025)	–	–	✓	✓	–
Shi et al. (2023b)	✓	–	✓	–	–
Qin et al. (2023)	✓	–	✓	–	–
Qi et al. (2025)	✓	–	✓	–	○
Golovneva et al. (2023)	–	–	✓	–	✓
Chu et al. (2024)	–	✓	✓	–	–
Mazzia et al. (2026)	✓	✓	–	✓	–
Ours	✓	✓	✓	✓	✓

Table 1: Prior work by multilinguality (Multi.), temporal focus (Temp.), CoT, distractors (Distr.) and trace-level analysis (Trace); ○ means partly covered.

The gap. As Table 1 shows, no study has measured how CoT, distraction and language interact on the same problems, or compared the structure of English and Hindi traces for them; TRD provides the problems but leaves CoT out.

4 Experimental Setup

Data. Figure 1 shows the pipeline. TRD has nine task types: date addition and subtraction, time addition and subtraction, date and time duration, date recurrence, interval date and day of week, which its authors group into three reasoning tasks (arithmetic, duration, recurrence) and two memorisation tasks (intervals, day of week). Five difficulty levels scale the offsets and spans involved, from *short* to *very very long*, and each question fills a carrier phrase written with native speakers, such as "Today is [DATE], what is the date going to be in [N] days?" (Mazzia et al., 2026). We extend the TRD generator so that, for each task and level, it draws 150 questions (seed 9), each with a no-distractor, a time-related and an unrelated version taken from TRD's templates, and replays the same random draws for English and Hindi. The two languages therefore share dates, offsets, gold answers and distractor choice, and differ only in wording.

Prompts and model. Both prompts share one template. The direct prompt says "Answer the following question directly. Do not provide any reasoning or explanation. Give only the final answer." The CoT prompt says "Solve the following question by reasoning through it step by step. Your reasoning must be written entirely in [lan-

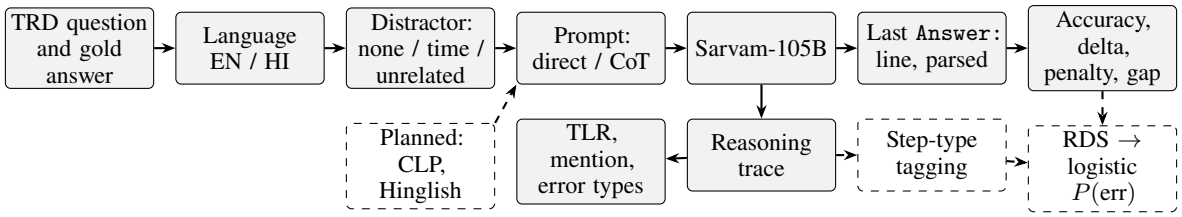


Figure 1: Our pipeline. Each TRD question is posed in English or Hindi, given a distractor, and wrapped in a direct or CoT prompt before Sarvam-105B answers it; the answer is scored and the trace analysed. Solid boxes are implemented and used in this report; dashed boxes are planned.

guage]; do not switch to another language at any point." Both then ask for the whole response in the question's language and for a final line of the form `Answer: <answer>` in a task-specific format, with the word `Answer:` kept in English and Western digits. We call `sarvam-105b`, a mixture-of-experts model built mainly for Indian languages, through Sarvam's API with greedy decoding (temperature 0, as in TRD), a 4,096-token output limit and the model's separate thinking mode turned off in both conditions, so that a direct answer really is direct and the CoT is the visible response.

Scoring and trace measures. We take the last `Answer:` line of a response, remove stray markdown and punctuation, convert Devanagari and Arabic-Indic digits to Western digits, and compare it with the gold answer by exact match, as TRD does; for day of week we compare the weekday itself, so that a Hindi or English spelling of the correct day both count. We measure language drift with the target-language ratio,

$$\mathrm{TLR}(t) = \frac{N_{\mathrm{hi}}(t)}{N_{\mathrm{hi}}(t) + N_{\mathrm{en}}(t)}, \qquad (1)$$

where N_{hi} and N_{en} count the Devanagari and Latin-script words in trace t outside the answer line; punctuation, symbols and digits count towards neither. For English traces the numerator counts English words instead. For Hinglish, where Hindi words are also written in Roman script, we will switch to word-level language identification.

5 Progress So Far

Pipeline. The solid boxes in Figure 1 are built and in use: the extended generator; an asynchronous runner that calls the Sarvam API in parallel, retries rate-limit and server errors with exponential back-off, and resumes an interrupted run without repeating finished requests; a scoring module for answer extraction, accuracy and the TLR; and analysis scripts for the measures and tests of Section 6. Every run is stored with a copy of its code, configuration and outputs.

Task	Short	Med.	Long	V. long	V.V. long
Date addition	150	150	150	150	–
Date subtraction	150	150	–	150	–
Date duration	150	150	135	150	–
Date recurrence	150	150	–	150	–
Day of week	146	149	–	150	–
Interval date	–	–	–	150	–
Time addition	–	–	–	149	–
Time subtraction	–	–	–	111	–
Time duration	–	–	–	149	–
Total	746	749	285	1309	–

Table 2: Questions answered in all 12 conditions, by task and difficulty level, out of 150 generated per cell. Cells marked – were not run, or only partly run, before the API credit ran out and are left out of the analysis.

Runs. Of the 6,750 questions planned (nine tasks, five difficulty levels, 150 questions each), 3,089 (46%) have been answered in all 12 conditions (Table 2). The runs stopped when the API credit ran out, so the very-very-long level has not been run and some tasks are missing at the short, medium and long levels. We analyse only questions with all 12 responses, which keeps every comparison paired.

Data checks. We checked that the English and Hindi versions are parallel. For all 3,089 questions the gold answer is identical across languages and all 12 records, the question numbers match across languages, the direct and CoT prompts share the same question, and every English distractor maps to exactly one Hindi distractor and back (35 time-related and 43 unrelated pairs).

6 Analysis

We analyse the 37,068 responses to the 3,089 questions answered in all 12 conditions (Section 5). All differences are in percentage points. We can

grade a response only if it ends with the requested `Answer:` line. 364 responses (1.0%) have no such line because they reached the 4,096-token output limit and were cut off mid-reasoning; 280 of them are English CoT. We leave these out of accuracy rather than count them as wrong, and check in Section 6.1 that counting them as wrong changes no conclusion.

Differences between paired conditions are tested with the exact McNemar test (McNemar, 1947) with Holm correction (Holm, 1979), and confidence intervals (CIs) come from a 2,000-sample bootstrap that resamples questions, so that the 12 responses to a question stay together (Berg-Kirkpatrick et al., 2012; Dror et al., 2018). Square brackets give 95% CIs, the range of values consistent with the data; an interval that excludes zero means the difference is unlikely to be due to chance.

6.1 Does CoT help, and in which language?

Table 3 shows the central result. CoT raises English accuracy from 63.2% to 91.1% (+27.9 points [+26.6, +29.3]) but Hindi accuracy only from 57.7% to 62.9% (+5.1 [+3.8, +6.4]). The English gain is significant in all 9 tasks. In Hindi, CoT helps significantly in 5 tasks and *hurts* significantly in two: date subtraction (92.2% → 78.2%) and interval date (35.1% → 17.9%). Asking the model to reason step by step in Hindi can therefore be worse than asking it for the answer alone, and this happens without any distractor, since the table pools all three distractor conditions. The only task where Hindi gains at least as much as English is time duration (+43.8 vs. +40.9 points), where Hindi direct answers are unusually poor (28.9%).

These conclusions do not depend on how the cut-off responses are treated. Counting them as wrong, which penalises English CoT most, the CoT gain is still +25.3 points in English and +4.7 in Hindi.

Direct answers on two tasks are at chance. Without CoT, day-of-week accuracy is 14.1% in English and 15.7% in Hindi, against a chance level of 14.3%. Figure 2 shows that this is not a near miss. The difference between the predicted and the correct weekday is spread evenly over all seven values (12.7–15.7% each), so the direct answer carries no information about the date. Instead, direct answers favour one weekday regardless of the question: Friday for 35.4% of English answers and Sunday for 63.4% of Hindi answers, while the correct

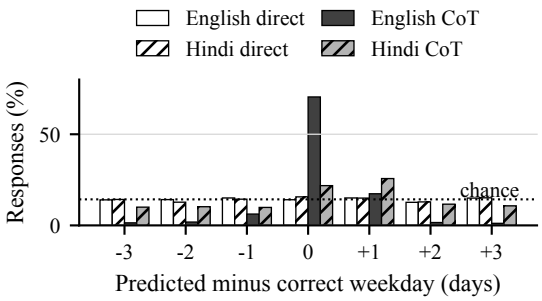


Figure 2: Predicted minus correct weekday for day-of-week questions (graded responses). Direct answers in both languages are spread evenly over all offsets, matching chance (dotted line, 1/7).

answers are spread roughly evenly over the week. English CoT fixes most of this (70.5%), with remaining errors skewed late (+1 day: 17.4%; −1 day: 6.2%). Hindi CoT is more often exactly one day late (25.7%) than correct (21.8%).

Date recurrence shows a similar split. In both languages, 31.3% of direct answers are "today plus one interval", which ignores which occurrence the question asks for. English CoT almost eliminates this shortcut (0.1%), while Hindi CoT keeps it in 16.6% of answers. Hindi also produces a date in the past, today minus the interval times the occurrence number, in 10.8% (direct) and 11.8% (CoT) of answers, against 1.1% and 0.5% in English. The Hindi template phrases the question as आज को छोड़कर, 1वीं आवृत्ति की तिथि क्या है? *āj ko choṛkar, 1vīṃ āvṛtti kī tithi kyā hai?* ('excluding today, what is the date of the 1th occurrence?'), with a templated ordinal suffix that is non-standard for small numbers. We have not yet tested whether this wording causes the reversal; it is a possible confound in TRD's Hindi recurrence questions rather than in the model's arithmetic.

6.2 The English–Hindi gap across difficulty

We call English accuracy minus Hindi accuracy the *English–Hindi gap*; a positive gap means English is ahead. Because each difficulty level contains a different mix of tasks, Figure 3 compares levels using only tasks present at all of them. From the short to the very long level, the gap grows far more with CoT than with direct answers. For date addition and duration it grows from 3.0 to 7.1 points with direct answers but from 6.6 to 18.5 points with CoT. Over the five date tasks, the CoT gap grows from 27.0 to 40.4 points, while the direct gap stays be-

		English			Hindi		
Task	Q	Direct	CoT	Δ	Direct	CoT	Δ
Date addition	600	93.7	98.5	+4.8*	89.5	88.3	−1.2
Date subtraction	450	93.6	95.9	+2.3*	92.2	78.2	**−14.0***
Date duration	585	88.9	99.1	+10.2*	85.1	87.5	+2.5
Date recurrence	450	24.5	92.4	+67.9*	4.8	26.0	+21.2*
Day of week	445	14.1	70.5	+56.4*	15.7	21.8	+6.1*
Interval date	150	29.8	43.1	+13.3*	35.1	17.9	**−17.2***
Time addition	149	57.7	97.1	+39.3*	60.2	73.5	+13.4*
Time subtraction	111	48.9	98.5	+49.5*	51.7	74.2	+22.5*
Time duration	149	58.8	99.8	+40.9*	28.9	72.6	+43.8*
All tasks	3,089	63.2	91.1	+27.9*	57.7	62.9	+5.1*

Table 3: Accuracy (%) with direct answers and chain-of-thought (CoT), pooled over difficulty levels and distractor conditions. Q is the number of questions; each question contributes three responses per language and condition. Δ is CoT minus direct in percentage points; * marks $p < 0.05$ in an exact McNemar test on paired responses, Holm-corrected over the 18 task–language tests (uncorrected for the pooled row). Significant negative deltas are in bold. Chance for day of week is 14.3%.

	Accuracy			Drop	
	None	Sim.	Dis.	Sim.	Dis.
English					
Direct	63.9	63.0	62.7	+0.9	+1.2
CoT	91.6	90.6	91.2	+0.9	+0.4
CoT − direct				+0.0	−0.8
Hindi					
Direct	57.4	58.1	57.7	−0.7	−0.3
CoT	65.2	60.7	62.7	+4.5	+2.5
CoT − direct				**+5.2**	**+2.8**

Table 4: Accuracy (%) with no distractor (None), a time-related (Sim.) or an unrelated (Dis.) distractor. Drop is accuracy without the distractor minus accuracy with it (points; positive means the distractor hurt). CoT − direct is the CoT drop minus the direct drop (the CoT distraction penalty); bold marks a 95% cluster-bootstrap interval that excludes zero.

low 9 points. On the easiest questions, Hindi CoT does not even beat Hindi direct answers (92.8% vs. 94.0%). The language gap is therefore mainly a gap in how much CoT helps.

6.3 CoT under distraction

Our main hypothesis was that CoT gives a distractor more opportunities to enter the reasoning and change the answer. Table 4 supports this only in Hindi. In English, neither distractor lowers accuracy by more than 1.2 points under either prompting condition. In Hindi, direct answers are unaffected by the similar distractor (accuracy changes by −0.7 points), but CoT accuracy falls by 4.5 points. CoT therefore loses 5.2 more points to the distractor than direct answering does [+2.9, +7.3]; we call this extra loss the *CoT distraction penalty* (the "CoT − direct" row of Table 4). In English the penalty is +0.0 [−1.7, +1.8]. The unrelated distractor also carries a smaller penalty in Hindi (+2.8 [+0.7, +4.9]). Computing the Hindi penalty for the similar distractor separately for each task shows that it comes mostly from date duration (+10.3 points [+5.3, +15.1]), day of week (+8.0 [+1.3, +14.2]) and time duration (+22.1 [+8.3, +35.5]). Previous work found that irrelevant context distracts models on arithmetic word problems (Shi et al., 2023a); here the susceptibility is specific to one language and to the reasoning condition.

Does the distractor enter the trace? We count a trace as mentioning the distractor if it contains a content word of the distractor that does not occur in the question (function words and generic time words such as "day" are excluded). Some of these words can appear in a trace by coincidence, for example month names in a trace about dates. To measure this chance rate we use a *placebo*: for each question we take the trace written without any distractor and search it for the words of the distractor that would have been inserted. The model never saw those words, so any match there is coincidence. The difference between the two rates is the share of traces that really take up the distractor. Hindi traces take up the distractor far more often than English ones (Table 5): 35.9% of Hindi traces

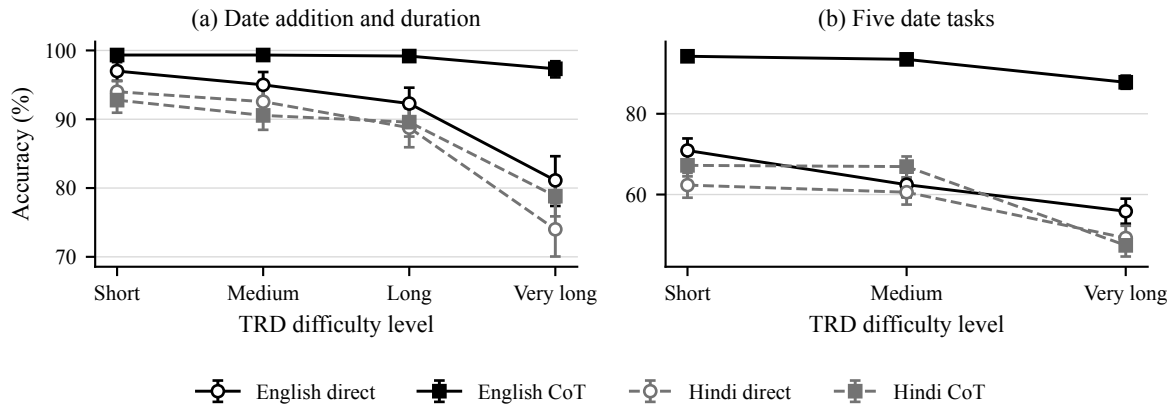


Figure 3: Accuracy by TRD difficulty level with the task set held fixed: (a) date addition and date duration, the only tasks present at all four levels; (b) the five date tasks present at the short, medium and very long levels. Black solid lines are English and grey dashed lines Hindi; filled squares are CoT and open circles direct answers. Error bars are 95% cluster-bootstrap CIs.

mention the unrelated distractor (placebo 1.4%), against 18.7% in English (placebo 8.5%). Hindi traces often build the distractor into the narrative of the solution. For the short date-addition question that starts "The garden needs new flowers", a correct Hindi trace ends with

> इस तरह, बगीचे में नए फूल लगाने के लिए सही तारीख 2025-03-30 है।
> *is tarah, bagīce meṃ nae phūl lagāne ke lie sahī tārīkh 2025-03-30 hai.*
> 'Thus, the correct date for planting new flowers in the garden is 2025-03-30.'

Appendix A.1 relates these mentions to answer correctness. The distractor also leaks into wrong answers. For example, after the distractor "February is such a unique month with its varying days", a Hindi direct answer to a recurrence question whose correct answer is 2025-04-10 was 2025-02-06. Overall, 12.1% (76 of 629) of wrong Hindi direct answers contain a number, month or weekday taken from the similar distractor. With the same placebo as above (wrong answers to the same questions asked without the distractor, checked against the distractor that would have been inserted), this happens in only 2.6% (16 of 607), so the difference is not chance (Fisher's exact test, $p < 0.001$). In English the two rates do not differ (4.9% vs. 3.9%, $p = 0.45$). In Hindi direct answers, the distractor therefore changes what the wrong answers are without changing how many there are.

6.4 Properties of the reasoning traces

The model stays in Hindi, and drifting into English helps. The target-language ratio (Table 5), the share of a trace's words that are in Hindi, is 0.986 for Hindi traces, and only 2.4% of them contain more than 10% English words (ratio below 0.9). We call these *drifting* traces. The Hindi results above therefore reflect reasoning written in Hindi. The 224 drifting traces look less accurate overall (37.5% vs. 63.5%), but mainly because of which questions they occur on: 181 of them are day-of-week questions, one of the two hardest tasks for Hindi CoT. Compared within the same task, drifting traces are more accurate: 34.3% vs. 19.8% on day of week and 51.4% vs. 25.3% on date recurrence. Combining all tasks this way, a drifting trace has about twice the odds of a correct answer (Appendix A.2). This agrees with the finding that English intermediate steps can match or beat steps

	English	Hindi
Cut off before answering	3.0%	0.7%
Median length (words)	103	79
Median completion tokens	327	198
Target-language ratio	1.000	0.986
Ratio below 0.9	0.0%	2.4%
Has a verification step	79.1%	1.5%
Answer value in trace	99.9%	99.0%
Mentions sim. distractor	15.1% (10.4)	31.8% (8.3)
Mentions dis. distractor	18.7% (8.5)	35.9% (1.4)

Table 5: Properties of CoT responses. All rows except the first are computed over graded responses. The target-language ratio is the number of target-language words divided by the number of Hindi plus English words, with both counts summed over all responses before dividing, so longer traces weigh more. Distractor mention rates are for traces whose prompt contained that distractor; the value in parentheses is the placebo rate for the same questions without it.

in the question's language (Shi et al., 2023b), and it motivates the cross-lingual prompting condition (Qin et al., 2023) that we run next.

Hindi traces are shorter and rarely check their work. Hindi traces are shorter than English ones (median 79 vs. 103 words; 198 vs. 327 completion tokens). 79.1% of English traces contain a verification step ("let me verify", "alternatively"), against 1.5% of Hindi traces. Verification does not explain the English advantage, however. English traces with a verification step look less accurate overall (89.5% vs. 97.4%), but again because verification is most common in the hardest tasks; compared within the same task, verifying makes no clear difference to accuracy (Appendix A.3).

Answers follow the trace. The final answer value appears in the trace in 99.9% of English and 99.0% of Hindi CoT responses, and only 1 of 798 wrong English answers and 6 of 3,416 wrong Hindi answers come from a trace whose last value was the correct one. Wrong answers are therefore wrong in the trace, not copied wrongly at the end. This is consistency between trace and answer, not evidence that the trace reflects the model's internal computation (Turpin et al., 2023).

6.5 Language-specific failures and error types

Every question was asked in both languages, so we can pair the English and Hindi responses to the same question, distractor and prompting condition and ask whether the two languages fail on the same questions. Under CoT, of 8,931 pairs, both are right in 5,565 and both wrong in 609; when they disagree, English is almost always the one that is right (2,585 pairs, against 172 where only Hindi is right). If Hindi failed mainly on questions that are hard in any language, the two languages would tend to fail together. We measure this with the ϕ coefficient, a correlation between two right/wrong outcomes (0: unrelated, 1: always the same), computed within each task and difficulty so that differences between tasks do not inflate it. With direct answers the median ϕ is 0.31: the languages partly fail on the same, harder questions. With CoT it is 0.05: whether Hindi CoT is right has almost nothing to do with whether English CoT is right on the same question. Hindi CoT failures are therefore specific to Hindi rather than a sign of hard questions.

Week	Work
5–11 Oct	Top up API credit; finish the missing short, medium and long runs (Table 2) and run the very-very-long level.
12–18 Oct	Hinglish reasoning condition and word-level language identification for its TLR; rerun Hindi date recurrence with a reworded question to test the template confound; start the rule-based step-type tagger.
19–25 Oct	CLP for Hindi questions, and CLSP with English and Hindi if time allows, compared with Hindi CoT; finish the step-type tagger and hand-tag a sample of traces to validate it and the mention, reuse and verification heuristics.
26–31 Oct	RDS and the logistic model $P(\text{err})$ from RDS, difficulty and language; significance tests across all conditions; final analysis, report and viva preparation.

Table 6: Timeline for the rest of the project (deadline 31 October).

The errors also differ in kind (Figure 4 in Appendix B). 66.5% of wrong English CoT answers are off by one day (88.9% in date addition and 85.5% in date subtraction), the slip of an otherwise correct procedure. Only 27.1% of wrong Hindi CoT answers are off by one. The rest include month slips, where the day and year are right but the month is wrong (17.5% of Hindi date-addition errors, none in English), and the reversed direction in date recurrence discussed in Section 6.1. Hindi CoT does not just make more of the same mistakes as English CoT; it often follows a different, wrong procedure.

7 Timeline and Remaining Work

Table 6 lists the remaining work. Completing the grid comes first; the new conditions follow in order of dependence, since Hinglish needs only a new instruction, CLP a second turn, and RDS a tagger validated by hand.

8 Conclusion

On 3,089 TRD questions answered in English and Hindi, with and without distractors, CoT helps Sarvam-105B far more in English than in Hindi, the gap grows with difficulty, and distractors harm CoT only in Hindi, where they often become part of the reasoning. The remaining work tests whether Hinglish reasoning or an English restatement closes this gap, and whether Hindi traces that diverge from English ones fail more.

Limitations

All results come from one model, decoded greedily with one prompt template, so they describe this configuration rather than CoT in general. Our runs stopped early when the API credit ran out, so the data cover only four of the five TRD difficulty levels and not every task at every level; Figure 3 works around this by comparing levels only on tasks present at all of them. Three of our measures, whether a trace mentions the distractor, whether a wrong answer reuses it, and whether a trace verifies its work, are simple word-matching rules rather than human judgements. They can miss a paraphrased mention and can count a word that appears for an unrelated reason. The placebo rates estimate how often such accidental matches happen, and comparing within each task removes differences between tasks, but neither catches missed paraphrases; we will check a hand-labelled sample for the final report. Cut-off responses are excluded from accuracy and are concentrated in English CoT, but counting them as wrong changes no conclusion. Finally, the CLP, Hinglish and trace-structure conditions are not yet run, so the claims here are limited to direct answers and CoT in the question's own language.

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A Odds-Ratio Analyses of CoT Traces

We relate three properties of CoT traces to answer correctness: mentioning the distractor, drifting into English, and containing a verification step. Pooled comparisons are misleading here, because tasks differ in both accuracy and how often each property occurs, and pooled comparisons can reverse (Simpson, 1951); this happens for drift and for verification. We therefore compare traces with and without the property within each task and difficulty, and combine these groups with a Mantel–Haenszel odds ratio (OR) (Mantel and Haenszel, 1959), using only groups that contain traces of both kinds. An OR of 1 means no difference; an OR above 1 means that traces with the property are more often correct, and an OR below 1 that they are less often correct. Square brackets give 95% CIs.

A.1 Distractor Mentions

Distractor	Language	Traces	OR [95% CI]
Time-related	English	2,974	0.64 [0.44, 0.93]
	Hindi	3,066	0.77 [0.63, 0.95]
Unrelated	English	3,015	1.33 [0.82, 2.15]
	Hindi	3,073	1.11 [0.90, 1.36]

Table 7: Odds ratio (OR) of a correct answer for CoT traces that mention the distractor versus traces that do not, stratified by task and difficulty (Mantel–Haenszel). Traces is the number of CoT responses whose prompt contained that distractor.

Table 7 compares CoT traces that mention the distractor (Section 6.3) with traces that do not. Traces that mention the time-related distractor have about 36% (English) and 23% (Hindi) lower odds of a correct answer, and both intervals lie below 1. For the unrelated distractor both intervals include 1, so mentioning it is not linked to accuracy: the model brings it up and then ignores it.

A.2 Language Drift

Section 6.4 compares drifting Hindi CoT traces (target-language ratio below 0.9) with the other Hindi CoT traces. Drifting traces are concentrated in the hardest tasks, so they look less accurate when pooled. Within task and difficulty, over the 12 groups that contain both kinds of trace, the OR is 1.94 [1.43, 2.64] ($p < 0.001$): a drifting trace has about twice the odds of a correct answer. This comparison is observational; drifting may accompany easier instances within a task rather than cause correct answers, which the cross-lingual prompting condition will test directly.

A.3 Verification Steps

English traces with a verification step look less accurate when pooled (89.5% vs. 97.4%), but verification is concentrated in the hardest tasks: it appears in 1,117 of the 1,141 graded English day-of-week traces. Within task and difficulty, the OR is 1.33 [0.90, 1.95] ($p = 0.13$). The interval includes 1, so verifying makes no clear difference to accuracy, and the much higher rate of verification in English (79.1% vs. 1.5%) does not explain the English advantage.

B Error Types

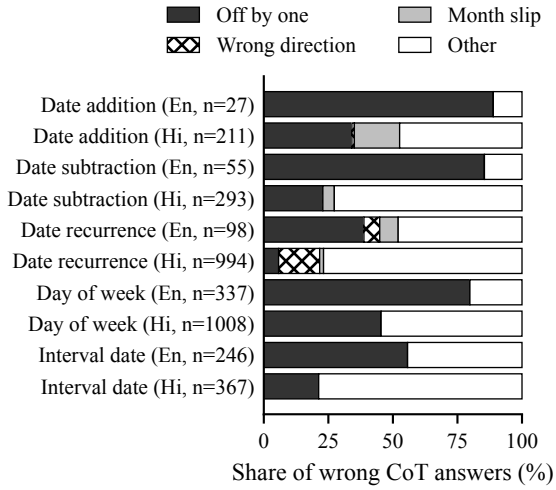


Figure 4: Composition of wrong CoT answers for tasks with at least 25 wrong answers in each language; n is the number of wrong answers. "Wrong direction" is the mirror image of the correct date around the date in the question.